Yiğit Özçelik

Computer Engineering Student · Robotics & Automation

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Profile

Second-year Computer Engineering student building software for robotics and automation, with a focus on autonomous ground vehicles: the software that lets perception, localization and planning run together on a real platform, backed by clear module interfaces, repeatable tests and simulation.

Competitions

TEKNOFEST 2026 Robotaxi Passenger Autonomous Vehicle

Ready-made vehicle category · Team AUtomotion-R · Finalist

- Started on the autonomous bus-stop manoeuvre (stop sign detection, bay entry, IMU-confirmed full stop, rejoining the lane), then grew to cover the architecture of the whole system.
- End-to-end stack on the BEE1 vehicle: CLRerNet lane detection, YOLO sign and light detection, Velodyne lidar obstacle zones, Xsens GNSS/INS localization, Lanelet2 and behaviour-tree planning, CAN control.
- ROS 2 Foxy in a single GPU Docker image, with a simulation bridge and scenario tests for each competition day.

Experience

National Technology Workshop (MTA)

Frontend development · Workshop tracking system

- Frontend of a progressive web app where students request materials and workshop staff manage approvals and hand-overs. React, TypeScript, TanStack, Tailwind CSS, Keycloak.

Active Projects · 2026

Unmanned Ground Vehicle (İKA) — Team captain

TEKNOFEST project. Part of the whole process as captain, owning the vehicle simulation: Blender model and Gazebo model generation in Python.

Advanced Autonomous Systems — Simulation, route planning

TEKNOFEST project. Infinite-seed Gazebo and RViz simulation: every seed is a new rubble field, scored in batch runs. ROS 2, Python.

Robotaxi: Original Vehicle — Simulation

Real-time CARLA simulation connected to the autonomy stack through a ROS 2 bridge.

Polar Research — Team captain

TÜBİTAK polar research project.

Swarm UAV

Active project on UAVs flying as a swarm.

Open Source

MTA Workshop Inventory — github.com/yigitozc/mta-atolye-takip

Material request and approval app in HTML, CSS and JavaScript.

Personal website — yigitozc.github.io

Library-free JavaScript simulation of lidar mapping, an occupancy grid and A* replanning.

Education

Ankara University

B.Sc. Computer Engineering

2025 – present

GPA 3.23

Skills

Programming

Python, C++, Java, JavaScript, TypeScript

Robotics

ROS 2, Gazebo, CARLA, RViz, YOLO, OpenCV, Lanelet2

Web

React, HTML, CSS, Tailwind CSS, Vite

Tools

Linux, Docker, Git, Blender, Maven, Playwright

Languages

Turkish

Native

English

Limited working proficiency

Japanese

Elementary proficiency

Certification

Programming with Python

MATHCOM, Ankara University · Nov 2025

Community

CORE Student Community

Substitute Board Member · Ankara University · 2025 – present